

Series III – Article III.6: Synthesis – Beal’s Conjecture Resolved (Updated – 33 Problems)

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May 2026

Abstract

We present a complete synthesis of the spectral proof of Beal’s conjecture, developed in Articles III.1–III.5. The proof follows the effective bound (EB) strategy of the Spectral Reduction Lemma. Beal’s conjecture is the third problem in the ordered list of **thirty-three resolved** by the spectral method (entry #3). We provide a correspondence dictionary linking the arithmetic of Beal to spectral concepts, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

Contents

1	Introduction	1
2	Recap of the four pillars for Beal	2
3	Correspondence dictionary: Beal spectral framework	2
4	Integration into the global corpus	2
5	Formalisation in Lean4	3
6	Conclusion	3

1 Introduction

Beal’s conjecture, formulated by Andrew Beal in 1993, states that the equation $A^x + B^y = C^z$ with $x, y, z > 2$ and $\gcd(A, B, C) = 1$ has no solutions. The spectral proof follows the effective bound (EB) pattern:

- Explicit effective bound N_0 (linear forms in logarithms) such that any counterexample satisfies $A, B, C \leq N_0$.
- Reduction to a finite SAT instance via a boolean circuit and the Tseitin transformation.
- Construction of the spectral Hamiltonian and verification of the four pillars.
- Application of the spectral SAT solver to prove unsatisfiability.

2 Recap of the four pillars for Beal

The spectral Hamiltonian H is built on the hypercube of assignments of the SAT instance $\Phi_{\text{Beal}}(N_0)$. The four pillars are verified exactly as in Series I (SAT):

1. **P1:** Unique strictly positive ground state ψ_0 .
2. **P2:** Constant spectral gap $\gamma(H) \geq 2$.
3. **P3:** Logarithmic area law, hence MPS representation with bond dimension polynomial in M .
4. **P4:** D-RSP algorithm extracts a satisfying assignment (or proves unsatisfiability).

3 Correspondence dictionary: Beal spectral framework

Beal concept	Spectral concept
Counterexample (A, B, C, x, y, z)	Configuration σ of bits
Equation $A^x + B^y = C^z$	Boolean circuit output 1
Explicit bound N_0	Parameter that fixes SAT instance size
SAT instance $\Phi_{\text{Beal}}(N_0)$	Potential $\hat{V}$ zero on satisfying assignments
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating all assignments
Constant spectral gap	Exponential convergence of power iteration
D-RSP algorithm	Deterministic extraction of a solution (or proof of unsatisfiability)

4 Integration into the global corpus

With Beal's conjecture proved, the list of thirty-three problems resolved by the spectral reduction lemma now includes it as entry #3.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap and confinement (II)	FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	FV
5	Goldbach (V)	CD
6	Kummer–Vandiver (VI)	FD
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
	– twin prime conjecture	CO
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
	– Hall (corollary of abc)	CO
14	Kithima–Landau (XIV)	FV
	– fourth Landau problem	CO
15	Hadamard matrices (XV)	CD

16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	FV
22	1/3–2/3 conjecture (XXII)	CD
23	Pillai (XXIII)	EB
24	n-conjecture (XXIV)	EB
25	Vizing (XXV)	CD
26	Erdős–Hajnal (XXVI)	EB
27	Gilbert–Pollak (XXVII)	FV
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Birch–Swinnerton-Dyer (BSD) (XXX)	SRP
31	Fuglede (XXXI)	SUTL
32	Barnette (XXXII)	SHS
33	Infinitude of prime families (XXXIII)	FV

5 Formalisation in Lean4

Listing 1: MainIII.lean (final)

```

1 import III.III1
2 import III.III2
3 import III.III3
4 import III.III4
5 import III.III5
6
7 theorem beal_conjecture :
8     (A B C x y z :      ), x > 2      y > 2      z > 2
9     Nat.gcd A (Nat.gcd B C) = 1      (A^x + B^y = C^z) :=
10     beal_spectral_proof

```

6 Conclusion

Beal’s conjecture is now unconditionally proved, adding a third major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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